

Back

Ideal Block Cipher

Graded Quiz • 20 min

Hide menu

- Modern Cipher and Module Objectives
- Modern Cipher and Block Cipher (vs. Stream Cipher)
- Ideal Block Cipher
- Video: Ideal Block Cipher7 min
- Quiz: Ideal Block CipherSubmitted
- Feistel Cipher
- Data Encryption Standard (DES)

Congratulations! You passed!

Grade received

100%

Latest Submission

Grade 100%

100% higher

Go to next item

Review Learning Objectives

1. You are given a stream cipher and a block cipher with a block size of 64 bits. The data input is 32 bits. How many bits do you need to pad before processing the data input using a stream cipher?

2 / 2 points

0

Submit your assignment

Due Apr 21, 11:59 PM IST

Attempts 2 every 8 hours

Try again

Correct

The stream cipher processes the data in bits (or the smallest unit for information processing). Therefore, it requires no padding.

Receive grade

To Pass 80% or higher

Your grade

100%

View Feedback

We keep your highest score

2. The block cipher supports input/output of 64 bits. Which of the following is true about the key storage requirement?

4 / 4 points

LikeDislikeReport an issue

- ☒ You need hundred million to one billion 1TB-hard-drives to store the key, where TB stands for Terabytes.
- ☐ You need more than one billion 1TB-hard-drives to store the key, where TB stands for Terabytes.
- ☐ You need ten to hundred million 1TB-hard-drives to store the key, where TB stands for Terabytes.
- ☐ You need less than ten million 1TB-hard-drives to store the key, where TB stands for Terabytes.

Correct

3. A block cipher takes a 3-bit plaintext and outputs a 3-bit ciphertext. If the cipher is an ideal block cipher, how many reversible transformations (keys) can it have?

1 / 1 point

40320

Correct

See Ideal Block Cipher lesson.